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RESEARCH ARTICLE

DM-UNet: Enhancing Spatial Awareness via Dynamic Mamba for Lightweight Brain Tumor Segmentation

CHENGCHENG JIN[1], NOR SAFIRA ELAINA MOHD NOOR[2], AND HAIDI IBRAHIM[3], (Senior Member, IEEE)

[1]School of Electrical and Electronic Engineering, Tunku Syed Sirajuddin, Engineering Campus, Universiti Sains Malaysia, Nibong Tebal, Penang 14300, Malaysia
[2]School of Electrical and Control Engineering, Ningxia Vocational Technical University of Industry and Commerce, Yinchuan 750000, China
[3]Department of Neurosciences, School of Medical Sciences, Health Campus, Universiti Sains Malaysia, Kubang Kerian, Kelantan 16150, Malaysia

Corresponding authors: Nor Safira Elaina Mohd Noor (safira@usm.my) and Haidi Ibrahim (haidi_ibrahim@ieee.org)

ABSTRACT Accurate brain tumor segmentation in Magnetic Resonance Imaging (MRI) is crucial for clinical diagnosis and treatment planning. Deep learning-based glioma segmentation can assist doctors in making more informed decisions. Some segmentation models are computationally intensive and memory-heavy, making them impractical for real-time deployment in resource-constrained environments. To further realize the clinical potential of these models, it is essential to ensure both high segmentation accuracy and lightweight design for efficient operations. However, these two goals are usually difficult to achieve simultaneously. Therefore, this paper proposes a Dynamic Mamba U-Net (DM-UNet) model, based on Mamba, fully leveraging the linear computational complexity of the Mamba architecture. Considering the morphological variability of brain tumors, both axial and coronal scan results are utilized, and a dynamic Mamba mechanism is designed. To capture shallow features, channel attention and dynamic Mamba are employed in a parallel manner. Partial convolution is integrated into multi-scale convolutional layers to obtain more detailed tumor features. With these approaches, the number of model parameters is reduced to less than 0.1M, the model size is merely 0.36M, and the computational cost is 14.96G FLOPs. We conduct 5-fold cross-validation on BraTS2023, obtaining an average Dice of 89.60% and an HD of 4.11 mm. The performance is superior to other advanced glioma segmentation models, providing further support for the deployment of lightweight models on edge devices.

INDEX TERMS 2D selective scanning (SS2D), channel attention dynamic mamba module (CADM), computer-aided diagnosis (CAD), convolutional neural networks (CNNs), cross-entropy (CE), deep learning (DL), dice similarity coefficient (DSC), dynamic tri-orientated spatial mamba (DTSM), enhancing tumor (ET), Hausdorff distance (HD), hierarchical decoupled convolution (HDC), magnetic resonance imaging (MRI), multi-layer perceptron (MLP), partial dilation convolution (PDConv), partially and dilated hierarchical decoupled convolution module (PDHD), selective state space (SSM), spatial shift convolution (SSC), squeeze-and-excitation (SE), tri-orientated spatial mamba (TSMamba), tumor core (TC), vision mamba (Vim), vision transformer (ViT), visual state space (VSS), whole tumor (WT).

I. INTRODUCTION

Glioma represents the most common type of primary intracranial tumor, with malignant gliomas being the leading cause of mortality among primary brain neoplasms [1], [2]. Magnetic Resonance Imaging (MRI) enables comprehensive assessment of brain tumor volume, providing objective and quantitative references to support surgical planning and clinical decision-making [3], [4]. Recently, Deep Learning (DL)-based Computer-Aided Diagnosis (CAD) systems have

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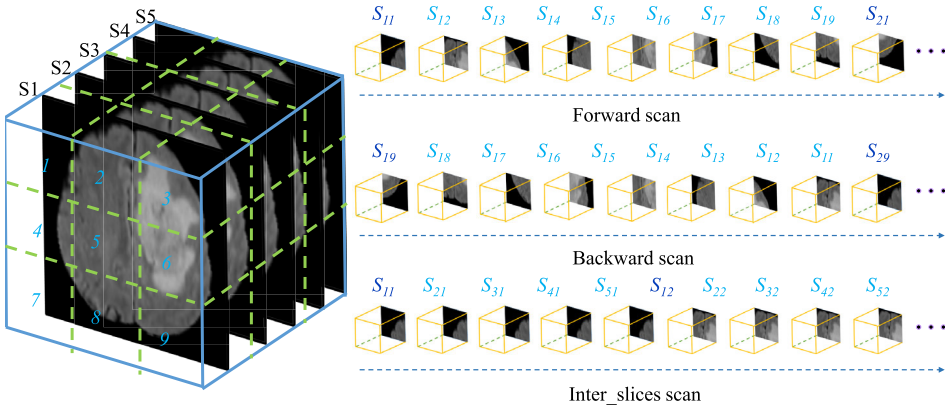
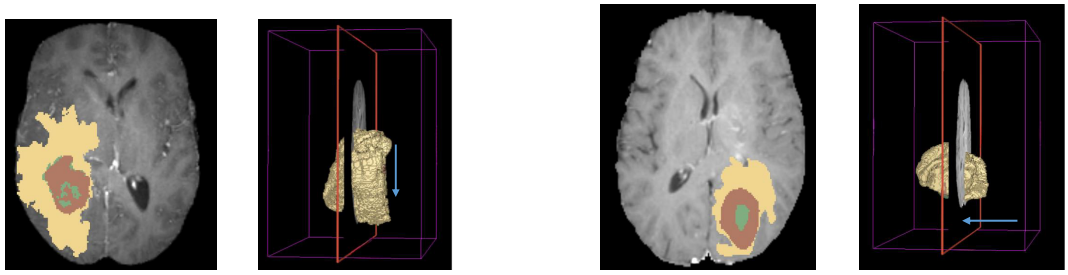


FIGURE 1. Tri-orientated spatial scan.



(a) BraTS-GLI-01418 (b) BraTS-GLI-01303

FIGURE 2. Gliomas with distinct morphologies.

demonstrated strong potential in automating glioma segmentation from MRI scans, facilitating early tumor detection and improving diagnostic accuracy [5].

To realize the clinical application of these DL models, researchers have increasingly focused on deploying them on compact and resource-constrained edge devices, such as the NVIDIA Jetson Xavier NX, Raspberry Pi, and Android-based mobile platforms. These devices offer advantages in portability, real-time inference, and cost-effective deployment. Liu et al. [6] proposed EDB-Diff, a diffusion-based model with only 15.1 million parameters, which is designed specifically for edge deployment on NVIDIA Jetson Xavier NX. Similarly, Memon et al. [7] applied post-training quantization to deploy the DeepLabV3+ model on Android smartphones for brain tumor classification and segmentation, reducing the model size from 172 MB to 43 MB. These applications highlight the potential of DL-based CAD models for clinical deployment via edge computing.

However, Memon et al. [7] observed a slight reduction in segmentation accuracy, following model quantization. Xia et al. [8] prioritized model lightweighting at the cost of segmentation performance, resulting in outcomes inferior to the baseline. Such trade-offs may introduce potential risks in medical applications. Therefore, achieving a balance between model efficiency and segmentation accuracy remains a critical objective for edge-deployable CAD systems in healthcare.

Fewer parameters, lower memory consumption, and faster inference speeds typically characterize lightweight models. The compact architectures and minimal redundancy make them more robust to post-training quantization, bringing stable and controllable performance degradation [9], [10]. Furthermore, with the strict computational and memory limitations of edge devices, lightweight models offer clear advantages in inference efficiency and deployment feasibility compared to conventional large-scale architectures [11]. Motivated by these factors, we aim to develop an efficient lightweight glioma segmentation model that achieves high segmentation accuracy with minimal parameter and memory overhead.

Researchers have explored Convolutional Neural Networks (CNNs) for automatic glioma segmentation. Among them, U-Net [12], one of the most influential architectures in medical image segmentation, as well as its lightweight variants [13], [14], [15], have achieved notable success in this task. It is primarily due to their symmetric encoder-decoder structure and strong ability to capture local features. However, this design may lead to information loss during

the down-sampling process. In contrast, the Transformer architecture [16], based on self-attention mechanisms, effectively models long-range dependencies among image patches, thereby offering superior performance in global feature extraction. To harness the complementary strengths of both paradigms, recent studies have integrated Transformer modules into U-Net-like frameworks to enhance segmentation accuracy [17], [18], [19]. However, the $O(n^2)$ complexity of self-attention poses a significant challenge for developing lightweight models.

Mamba, proposed by Gu and Dao [20], is a novel sequence modeling framework with linear computational complexity $O(n)$, enabling substantial reductions in parameter count and computational overhead. These properties make it suitable for edge deployment. Similar to Transformer, Mamba processes input data as sequences. Expanding on this, Vision Mamba [21] performed bidirectional (forward and backward) scans over image patches to model spatial dependencies. While it is effective for 2D tasks, such bidirectional processing may fail to capture inter-slice continuity in 3D medical imaging, particularly in volumetric MRI-based tumor segmentation. To overcome this limitation, Xing et al. [22] introduced the Tri-orientated Spatial Mamba (TSMamba) block in the SegMamba model, which adds a third scanning direction across slices, as illustrated in Figure 1. This design enhances inter-slice information flow and enables more comprehensive tumor feature preservation in 3D brain MRI segmentation.

Nonetheless, gliomas vary greatly in morphology, size, and location across patients. Figure 2 illustrates two cases from the BraTS2023 dataset: BraTS-GLI-01418 and BraTS-GLI-01303, each with a 2D axial view (left) and a 3D segmentation (right). These images present the ground truth of glioma segmentation results, visualized using 3D Slicer. Yellow, dark brown, and green correspond to the Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET), respectively, representing different tumor subregions. The blue arrows indicate the predominant direction of tumor proliferation.

In case (b), the tumor extends more along the coronal plane, while case (a) has a larger volume with greater elongation in the axial direction. To improve segmentation accuracy, we focus on extracting features along the most informative directions. When infiltration is deeper across slices, coronal-plane scanning captures richer structural details; for tumors spreading axially, bidirectional in-plane scanning helps delineate clearer boundaries. To adaptively capture such variations, we propose the Dynamic Tri-orientated Spatial Mamba (DTSM), an extension of TSMamba to better model 3D tumor structures.

Based on this, we incorporate the DTSM module into the U-Net, proposing a Dynamic Mamba U-Net (DM-UNet) model. Notably, the model integrates an attention mechanism to Mamba for dynamically selecting the optimal directional features, allowing it to adapt to structural variations in brain tumors. In addition, partial convolution and dilated convolution are employed to capture the tumor details. Experiments on the BraTS2023 dataset demonstrate that our model significantly reduces the number of parameters while achieving superior performance, particularly in segmenting the challenging ET region. These results highlight the model's potential for deployment in resource-limited settings.

Our main contributions are summarized as follows:

- We propose DM-UNet, a lightweight glioma segmentation framework that integrates the Mamba module into a U-Net architecture, with fewer than 0.1M parameters, a size of only 0.36 MB, and a computational cost of 14.96G FLOPs.
- We design a DTSM module to effectively extract glioma features with diverse morphological characteristics.
- We fuse DTSM with a channel attention mechanism, enabling the model to capture fine-grained tumor details while significantly reducing the number of parameters.
- We combine partial convolution and dilated convolution to enhance multi-scale information extraction.

The remainder of this paper is organized as follows. Section II reviews related work on lightweight brain tumor segmentation in DL. Section III describes the proposed network architecture, while Section IV presents the implementation details, performance evaluation, and corresponding discussions. Finally, Section V concludes the paper and outlines future research directions.

II. PRIOR LIGHTWEIGHT MODELS

In this section, we first review recent studies on lightweight brain tumor segmentation models, focusing on approaches based on CNN and Transformer. As our proposed method is a novel glioma segmentation model based on the Mamba architecture, we then introduce the underlying principles of Vision Mamba, followed by a review of lightweight Mamba variants.

A. CNN-BASED AND TRANSFORMER-BASED WORKS

Depthwise separable convolutions provide an efficient means of reducing model complexity and parameter count. It involves two stages: each input channel undergoes an independent convolution, also known as channel-wise, which is followed by a point-wise 1×1 convolution to integrate information across channels. It maintains representation capacity and promotes lightweight design, benefiting constrained environments. On this basis, Howard et al. [27] constructed MobileNet for tasks such as object detection and fine-grained classification on mobile and embedded devices. Subsequently, Sandler et al. [28] further extended it by adding inverted residuals before depthwise separable convolutions. They constructed Mobile DeepLabv3, which is a lightweight model tailored for semantic segmentation on edge devices. However, relying solely on depthwise separable convolutions may limit feature representation and compromise segmentation performance [26]. Therefore, Kuang et al. [23] proposed a multi-path convolutional module

TABLE 1. Some lightweight 3D models based on CNN and Transformer for brain tumor segmentation.

Models	Dataset	Para (M)	FLOPs (G)	Dice (%)				Architecture	Highlights
				WT	TC	ET	Mean		
LW-CTrans [23]	BraTS2019	2.21	113.00	91.22	84.87	75.17	83.75	Hybrid	Adopt multi-path convolutional modules with directional convolution.
WIAF [24]	BraTS2020	5.23	9.75	90.50	87.80	76.80	85.00	Hybrid	Utilize iterative axial factorization attention to reduce the computational cost.
ADHDC-Net [13]	BraTS2020	0.30	25.81	89.91	83.12	77.85	83.63	CNN	Incorporate dilated convolution into hierarchical decoupled convolution.
TDPC-Net [14]	BraTS2020	0.19	17.96	90.01	83.43	78.18	83.87	CNN	Adopt partial convolution into hierarchical decoupled convolution to reduce the parameters.
GCI-Net [25]	BraTS2021	13.36	50.17	92.80	91.69	89.38	91.29	Hybrid	Introduce 3D convolutional attention to capture long-range dependencies.
LR-Net [26]	BraTS2021	4.72	990.91	92.80	90.60	85.10	89.50	CNN	Use the shift operation to expand the receptive field with reduced cost.

by performing deep convolution in different directions. As an alternative to depthwise separable convolutions, it leverages the anisotropic morphological characteristics of gliomas.

On the other hand, the lightweight models based on CNN mainly focus on the variants of U-Net, such as ADHDC-Net [13] and TDPC-Net [14]. The parameters are 0.30M and 0.19M, respectively, both benefiting from the hierarchical decoupled convolution. The former incorporates dilated convolution to capture multi-scale tumor features, while the latter adopts partial convolution to further reduce model complexity. Despite their lightweight nature, the segmentation performance of these models remains far from optimal, indicating substantial potential for improvement, particularly in capturing the complexity of brain tumor structures.

Transformer-based architectures for glioma segmentation face inherent challenges in lightweight design, primarily due to the computational burden of self-attention mechanisms. To mitigate this, Zhong et al. [24] proposed an iterative axial attention mechanism that 3D spatial dependencies along individual axes, significantly lowering computational cost while maintaining sensitivity to long-range spatial structures. Similarly, Qiao et al. [25] incorporated a 3D multi-scale convolutional attention module into the encoder to capture richer spatial features of brain tumors. This approach also enhances information fusion across skip connections via global context refinement. Meng et al. [26] introduced the Spatial Shift Convolution (SSC) module, which achieves broader receptive fields via kernel shifting. Even with small convolution kernels, this method can achieve a larger receptive field without increasing the number of parameters.

As summarized in Table 1, we provide an overview of representative lightweight models based on CNNs, Transformers, and hybrid architectures. The table focuses on key characteristics, including parameter count, computational complexity, segmentation accuracy, and core lightweighting strategies. It facilitates a clear comparison of the effectiveness of different approaches in lightweight model design.

B. MAMBA-BASED WORKS

Mamba achieves linear complexity via hardware-aware state expansion and selection mechanisms. The core of the Mamba architecture is the Selective State Space (SSM) mechanism, which adaptively retains relevant information while filtering out redundancy based on input content. Similar to the self-attention mechanism, this selective processing enhances contextual modeling by focusing on salient features. Hence, Mamba effectively captures long-range dependencies, boosting performance in deep contextual understanding tasks such as natural language processing.

Motivated by Mamba's efficiency in lightweight modeling, researchers have extended its application to vision tasks. Zhu et al. [21] developed Vision Mamba (Vim), which builds upon the original Mamba by incorporating bidirectional scanning. Specifically, Vim employs Mamba blocks that process images in both forward and backward directions to enhance contextual representation. The structural details are illustrated in Figure 3a. Similar to the Vision Transformer (ViT) [29], Vim also divides the input image into a series of patches and linearly projects them into vectors. The image patches are regarded as sequence data, and the visual representation is compressed by a bidirectional SSM. Moreover, with position embeddings, Vim gains spatial perception ability, enhancing robustness in dense prediction tasks.

The core process of Vim includes five parts: making patches, patch embedding, position encoding and position embedding, Vision Mamba encoder, and Multi-Layer

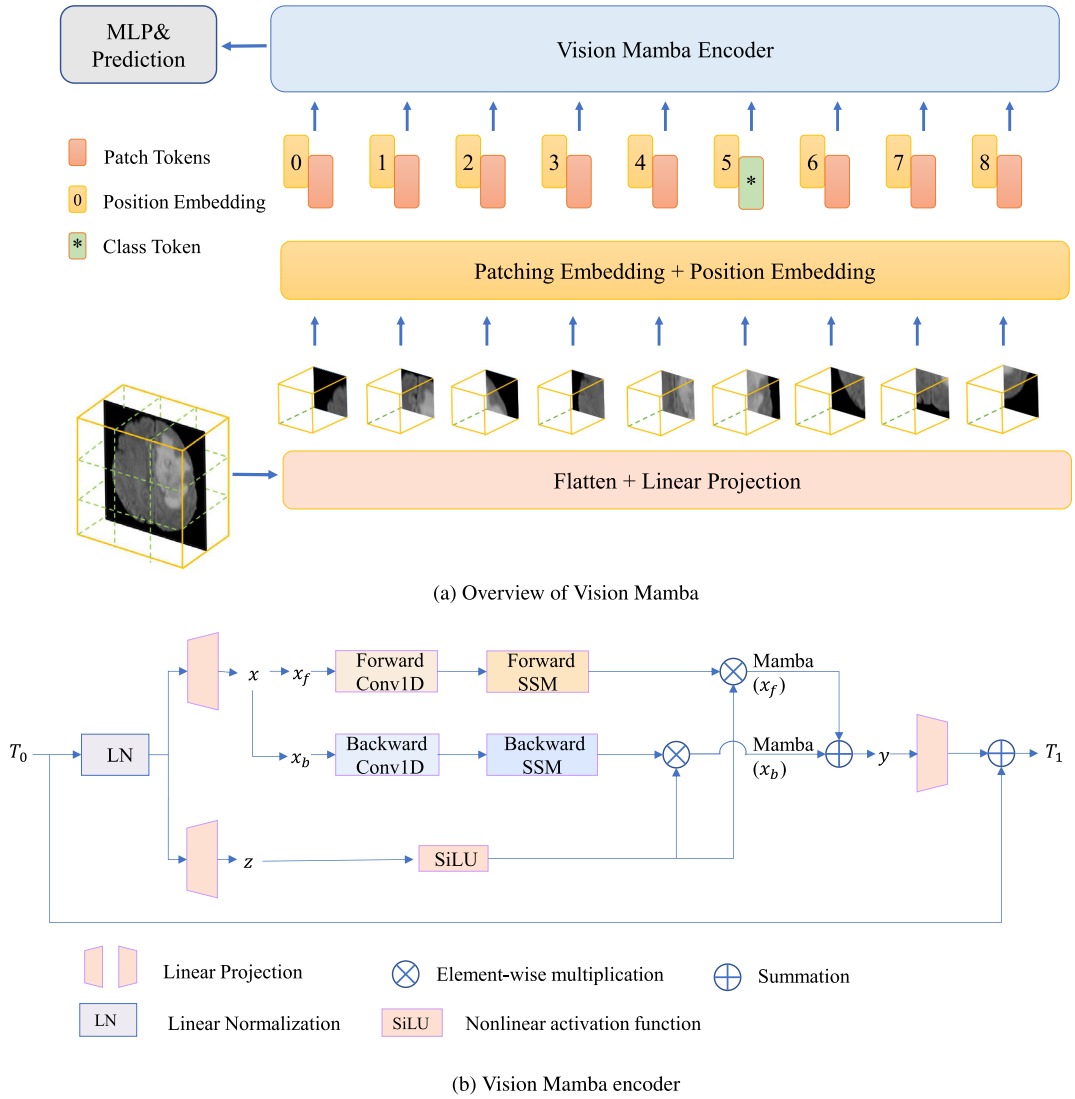


FIGURE 3. Overview of vision mamba and its encoder structure.

Perceptron (MLP) classification and prediction processing [21]. Consider a 2D image as an input, the first step is image preprocessing and patch generating. The image $x \in \mathbb{R}^{C\times H\times W}$ is split into a sequence of non-overlapping patches $\{x_1, x_2, \ldots, x_N\}$, wherein $x_i \in \mathbb{R}^{C\times P\times P}$, C is the number of channels, $[H, W]$ denotes the image size, $[P, P]$ is the resolution of a patch, and $N = \frac{HW}{P^2}$ is the number of patches. Next is flattening and linear projection, that is, flattening each patch, and then a linear projection is conducted to obtain the token, as shown in Eq. (1), which remains constant throughout the Vision Mamba.

$$x' = \{x_1\, E, x_2\, E, \ldots, x_N E\},\ E \in \mathbb{R}^{CP^2\times D} \tag{1}$$

where x' is the token, E is the learnable projection matrix, and D is the embedding dimension.

Third, position encoding and positional embedding. To maintain the spatial position information between the input image patches, the standard learnable 1D position embedding E_{pos} is embedded in the token, as shown in Eq. (2). In addition, the class token is used to represent the whole patch sequence, which is denoted as x_{class}.

$$T_0 = \{x_{\text{class}}; x'\} + E_{\text{pos}},\ E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D} \tag{2}$$

Then, the resulting tokens T_0 are fed into a Vim encoder to get T_1, as shown in Figure 3b. These tokens are split into two branches: the main feature branch x, learning the feature representation in two directions, and the gated branch z, introducing nonlinear features through the SiLU function. In the x branch, the data is first extracted by Conv1D for local features, followed by SSM. The process is expressed by the following formula:

$$\begin{aligned} T_1 &= Linear(y) + T_0, \\ y &= Mamba(x_f) + Mamba(x_b), \\ Mamba(x_f) &= SSM\left(SiLU(Conv1D(x_f))\right) \otimes SiLU(z), \\ Mamba(x_b) &= SSM\left(SiLU(Conv1D(x_b))\right) \otimes SiLU(z). \end{aligned} \tag{3}$$

where T_1 is the output token sequence and y represents the sum of the Mamba operations. *Linear* denotes a fully connected (linear) layer that projects the feature dimensions. x_f, x_b represent the flattened sequences of the input image after normalization and linear projection, which are along the forward and backward directions, respectively. $Mamba(x_f)$ and $Mamba(x_b)$ are the outputs of forward and backward Mamba operations, respectively.

Next, iterate q layers Mamba in a loop, wherein Vim follows the same logic as that T_1 obtained in the aforementioned process. The token sequence T_{q-1} is sent to the q-th layer of the Vim encoder, with a residual connection added to generate the output T_q. After normalization, the class token T_q^0 is extracted and subsequently fed into an MLP head to produce the final prediction $\hat{p}$, illustrated as follows:

$$\begin{aligned} T_q &= Vim(T_{q-1}) + T_{q-1}, \\ f &= Norm(T_q^0), \\ \hat{p} &= \text{MLP}(f). \end{aligned} \tag{4}$$

where *Vim* is the Vision Mamba block and *Norm* denotes the normalization layer.

Inspired by Vim, an increasing number of Mamba-based frameworks have emerged in medical image segmentation. Recent efforts focus on integrating Mamba with U-Net architectures to strengthen sequence modeling capabilities. Representative works include U-Mamba [30] and Seg-Mamba [22], both of which adopted a U-shaped architecture. These approaches differ from the classic U-Net by replacing its encoder and decoder with pure SSM modules or hybrid blocks that combine CNNs with SSMs.

Furthermore, VMamba [31] proposed a 2D Selective Scanning (SS2D) module, grounded in the Visual State Space (VSS) framework. By scanning along four distinct directional paths, the module aggregates contextual information from multiple perspectives. This process converts non-causal visual inputs into structured patch sequences, enhancing perception of complex scenes. Based on this paradigm, VM-Unet [32] and its improved variant VM-Unet-V2 [33] substituted the Transformer modules with VSS and incorporated them into the U-Net framework to enhance segmentation performance.

However, these models have significantly more parameters than the U-Net and its variants. In response, some work has shifted toward lightweight architectures. Notably, LightM-UNet [34] offers a compact solution by integrating pure Mamba modules into the U-Net architecture, and employs a residual Vision Mamba layer to capture deep semantic features. Impressively, this design achieves strong segmentation performance with only 1 million parameters. UltraLight VM-UNet [35] processes visual Mamba features in parallel, achieving a parameter count as low as 0.049M. SMM-UNet [36] further reduced parameters to 0.038M by dynamically adjusting the fusion strategy for local and global features, and capturing multi-scale information from varying sizes of lesions. Notably, these two models are designed exclusively for 2D image processing. Despite its strength in global context modeling, Mamba remains suboptimal at capturing fine-grained local details.

III. METHODOLOGY

To address the aforementioned limitations, we propose a lightweight glioma MRI segmentation model based on dynamic Mamba. Subsection III-A presents the overall architecture of our model. Then, Subsection III-B introduces the DTSM, and Subsection III-C provides detailed information about the Channel Attention Dynamic Mamba module (CADM). Subsection III-D elaborates the specific details of the Partially and Dilated Hierarchical Decoupled convolution module (PDHD). Finally, the loss function adopted for training is presented in Subsection III-E.

A. MODEL ARCHITECTURE

The proposed DM-UNet architecture is shown in Figure 4. Similar to most medical image segmentation models, DM-UNet retains a symmetric structure and skip connections similar to U-Net. Firstly, the multimodal MRI data undergoes preliminary feature extraction through depthwise separable convolution. To minimize the model parameters and computational requirements while capturing the detailed features and edge boundaries of brain tumors, we introduce two types of modules in the encoder part. In the first two stages, the CADM is introduced. Next, the PDHD module is adopted in the following two stages. In the first layer of the decoder, PDHD is still employed to extract the deep tumor detail feature. Then, in the other two decoder layers, the Hierarchical Decoupled Convolution (HDC) [37] unit is adopted. Finally, after normalization and activation, a depthwise separable convolution is applied to produce the final segmentation result.

B. DYNAMIC TRI-ORIENTATED SPATIAL MAMBA MODULE

In the DTSM block, we employ a dynamic tri-orientated spatial modeling strategy based on TSMamba, as shown in Figure 5. It includes forward scanning, backward scanning, and inter-slice scanning. The first two are performed in the axial direction, capturing in-plane spatial dependencies, while the latter operates along the coronal direction to enhance contextual interaction between adjacent slices. The most crucial difference between DTSM and TSMamba lies in the feature aggregation mechanism. TSMamba uses fixed and direction-independent weights to aggregate features, which may not sufficiently capture the heterogeneous morphology and spatial distribution of gliomas. In contrast, DTSM introduces an adaptive weight mechanism based on Squeeze-and-Excitation (SE) in the sequential modeling process of the axial and coronal directions. This mechanism dynamically adjusts the contribution weights of different directions according to the input tumor features and highlights the most relevant spatial information for each tumor structure to better represent brain tumor features. Specifically, this module learns image features from the main feature branch,

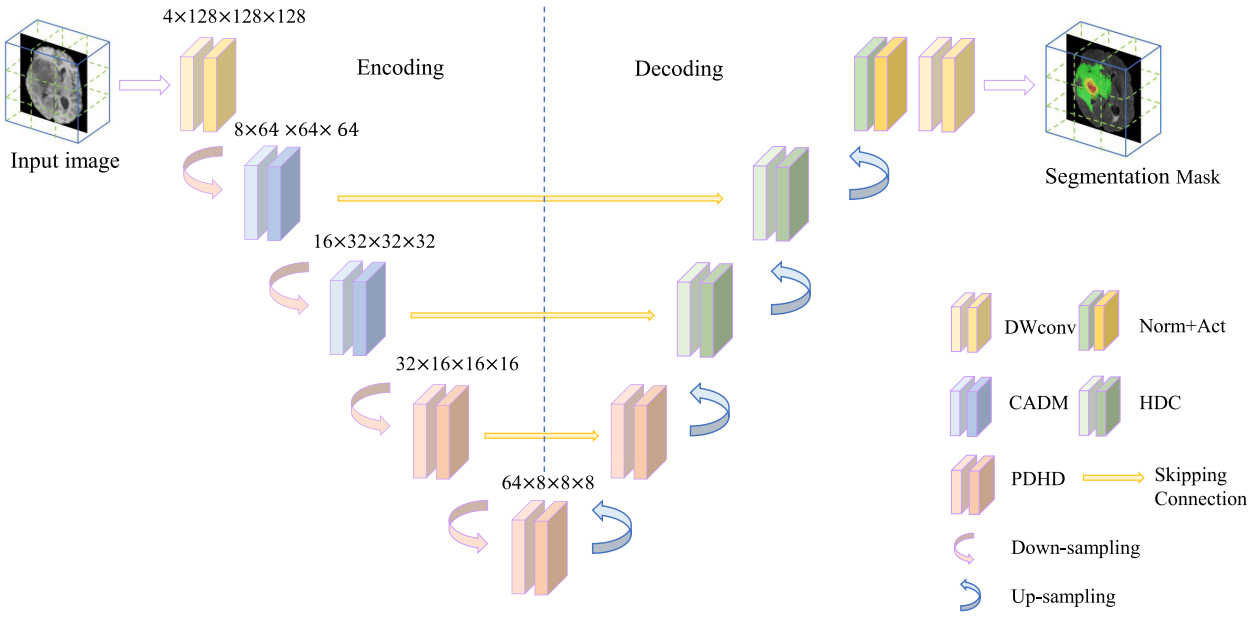


FIGURE 4. The architecture of the proposed DM-UNet.

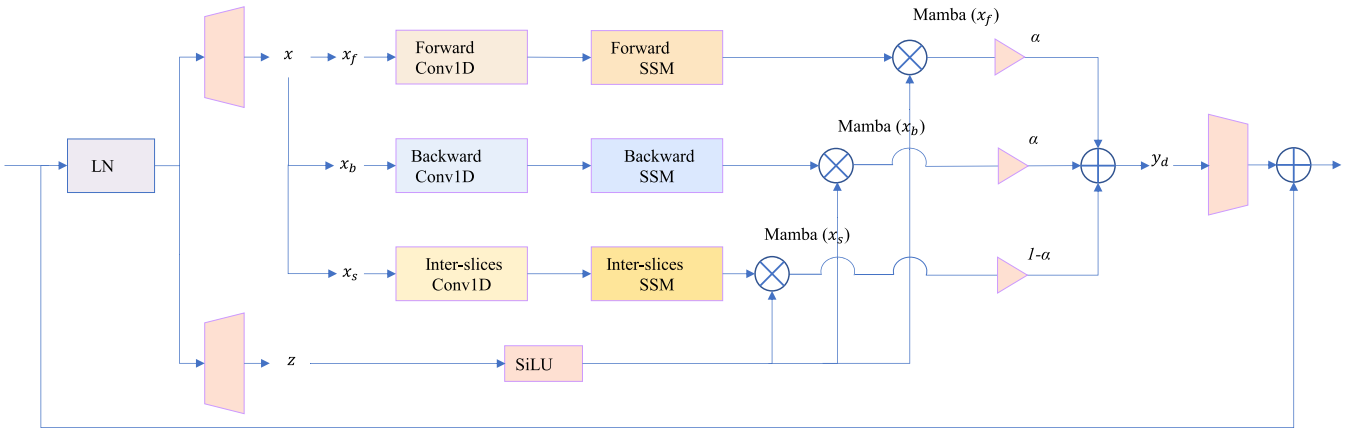


FIGURE 5. Dynamic Tri-oriented Spatial Mamba (DTSM) Block.

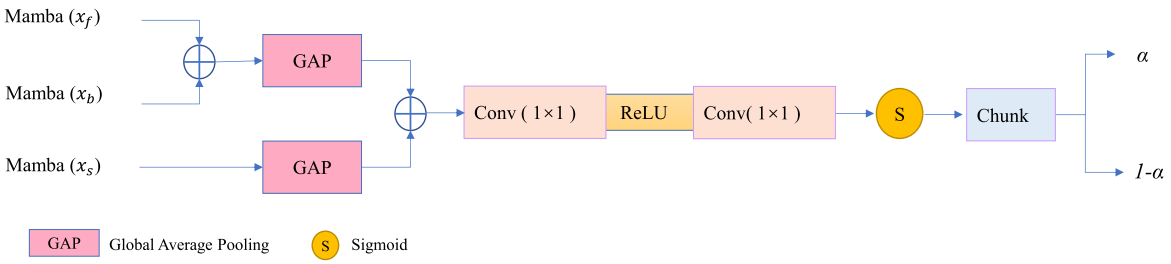


FIGURE 6. SE_LightAttention for dynamic axial-coronal feature reweighting using squeeze-and-excitation.

consisting of the three-directional information. Notably, we combine SE_LightAttention to give weights of α, α, and $1-\alpha$, respectively. The dynamic Mamba is calculated using the following Eq. (5).

$$y_d = \alpha(Mamba(x_f) + Mamba(x_b)) + (1-\alpha)Mamba(x_s) \quad (5)$$

where y_d denotes the sum of the dynamic Mamba operation. $Mamba(x_f)$, $Mamba(x_b)$, and $Mamba(x_s)$ represent the Mamba scanning results in different directions, whose calculation method is as shown in Eq. (3). α is the weighting coefficient for axial scanning.

The process of calculating weights is illustrated in Figure 6. It effectively adapts to the diversity of tumor shapes without

introducing additional computational overhead. First, the sum of the axial output, Mamba(x_f) and Mamba(x_b), is performed for Global Averaging Pooling (GAP). Similarly, the same operation is conducted on the coronal output Mamba(x_s). Then the two results are summed to obtain the fused global information GAP(x). Next, apply squeeze and excitation mechanisms to the aggregated feature representation, using two 1×1 convolutions with an intermediate ReLU for nonlinearity. The first convolution compresses the channels, the second expands them, followed by a sigmoid to rescale activations to [0, 1]. Finally, divide the output into two channel-wise components to obtain the attention weights, wherein α acts on the axial direction and $1-\alpha$ acts on the coronal direction. The SE_LightAttention dynamically adjusts the weights of axial and coronal features based on global information, selectively amplifying beneficial feature weights, hence enhancing the efficacy of feature fusion. The formula is as follows.

$$\begin{aligned}\mathrm{GAP}(x) &= \mathrm{GAP}\big(\mathrm{Mamba}(x_f)+\mathrm{Mamba}(x_b)\big)\\ &\quad+\mathrm{GAP}\big(\mathrm{Mamba}(x_s)\big),\\ (\alpha,1-\alpha) &= \mathrm{Chunk}\Big(\mathrm{Sigmoid}\Big(\mathrm{Conv}_{1\times1}\Big(\\ &\quad\mathrm{ReLU}\big(\mathrm{Conv}_{1\times1}(\mathrm{GAP}(x))\big)\Big)\Big)\Big)\end{aligned}\tag{6}$$

C. CHANNEL ATTENTION DYNAMIC MAMBA MODULE

To capture tumor features more accurately, we design the CADM module, as shown in Figure 7a. Cheng et al. [38] integrated Vim with a channel attention mechanism within a MetaFormer-style model, improving the image super-resolution performance. Inspired by this model, we incorporated DTSM into the MetaFormer architecture, which comprises two layers of normalization, the parallel DTSM block, the channel attention block, and the MLP block. Thus, CADM utilizes DTSM to extract global information while employing the channel attention mechanism to activate more useful pixels to supplement the global information. Its structure is relatively simple, and the processing of the CADM is formulated as follows:

$$\begin{aligned}y &= y_2+x,\\ y_2 &= MLP(LN(y_1))+y_1,\\ y_1 &= CAB(LN(x))+DTSM(LN(x))+x\end{aligned}\tag{7}$$

where y denotes the output image, x is the input image, LN denotes the layer normalizationand, CAB is the channel attention block for 3D.

The structure of CAB is shown in Figure 7b. The channel attention 3D mechanism improves the perception of important channel features. First, local features are extracted using two $3\times3\times3$ convolutions with GELU activation. Then, the channel attention module summarizes the features of each channel through GAP, followed by two $1\times1\times1$ convolutions to adjust channel dimensions. ReLU introduces nonlinearity, and Sigmoid generates channel weights, which are multiplied by the original feature map to enhance important features and suppress irrelevant ones. The CAB mechanism is depicted by the following equation:

$$\begin{aligned}x_{out} &= x_2,\\ x_2 &= \mathrm{Sigmoid}\Big(\mathrm{Conv}_{1\times1\times1}\big(\mathrm{ReLU}(\mathrm{Conv}_{1\times1\times1}(\\ &\quad\times\mathrm{GAP}(x_1))\big)\Big)\times x_1,\\ x_1 &= \mathrm{Conv}_{3\times3\times3}\Big(\mathrm{GELU}(\mathrm{Conv}_{3\times3\times3}(x_{in}))\Big)\end{aligned}\tag{8}$$

where x_{out} is the output feature and x_{in} denotes the input features.

D. PARTIALLY AND DILATED HIERARCHICAL DECOUPLED CONVOLUTION MODULE

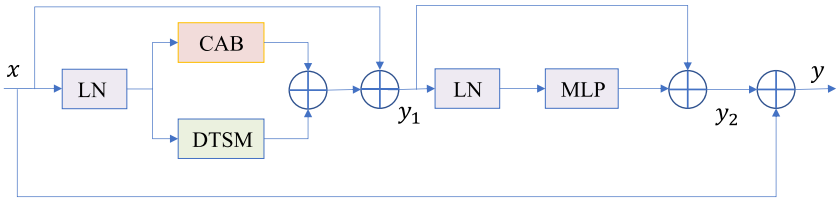
The HDC module, as illustrated in Figure 8a, first divides the input feature image into four parallel branches along the channel. Except for the first branch retaining the original channel information, the next three branches extract the features within the slices through $3\times3\times1$ convolution.

Notably, in order to enhance feature reuse and inter-branch information flow, a cascade strategy is implemented across these three branches. Specifically, the features of the preceding channel are transferred to the current channel. After concatenating the output from the four branches, the features between the tumor slices are extracted using $1\times3\times3$ convolution. This design not only diminishes the parameters, but also improves the feature perception ability of brain tumor 3D MRI. In addition, the asymmetric feature extraction framework provides the model with enhanced flexibility.

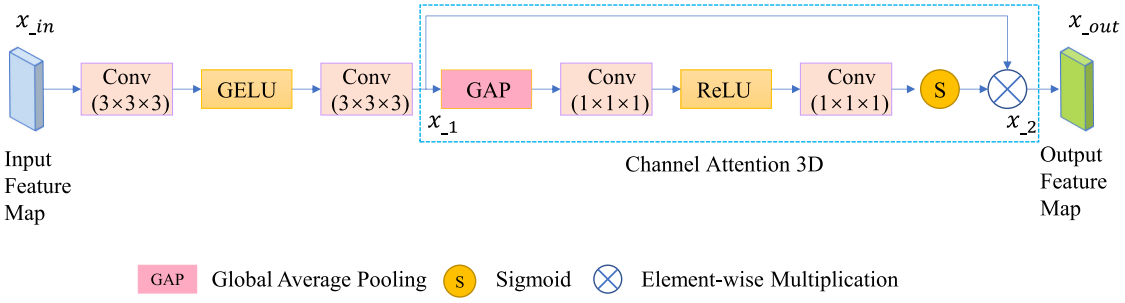
However, this module also has certain limitations. While reducing the complexity of the model, it limits the receptive field of features and hinders tumor detail recognition, indicating significant potential for enhancing segmentation accuracy in subregions for ET and TC. Inspired by the HDC and partially hierarchical decoupling convolution unit from [14], we developed the PDHD module, as shown in Figure 8b.

In contrast to the HDC, the crucial improvement involves substituting the $3\times3\times1$ convolution with Partial Dilation Convolution (PDConv). Additionally, the $1\times3\times3$ convolution has been removed. The dilated convolution provides multiple-scale receptive fields for superposition, facilitating the acquisition of more comprehensive information regarding brain tumors. The partial convolution is weighted adaptively through channel splitting. PDConv decreases the parameter and enhances the adaptability for various tumor sizes and shapes. In the three branches, PDConv utilizes dilation rates that range from small to large to mitigate feature information loss associated with the checkerboard effect [39]. The process of PDHD is represented by the following formula.

$$\begin{aligned}y &= Conv_{1\times1\times1}(x_{Cat})+x,\\ x_1,x_2,x_3,x_4 &= Conv_{1\times1\times1}(x),\\ x_{Cat} &= x_1+x_{2_PD}+x_{3_PD}+x_{4_PD},\end{aligned}$$

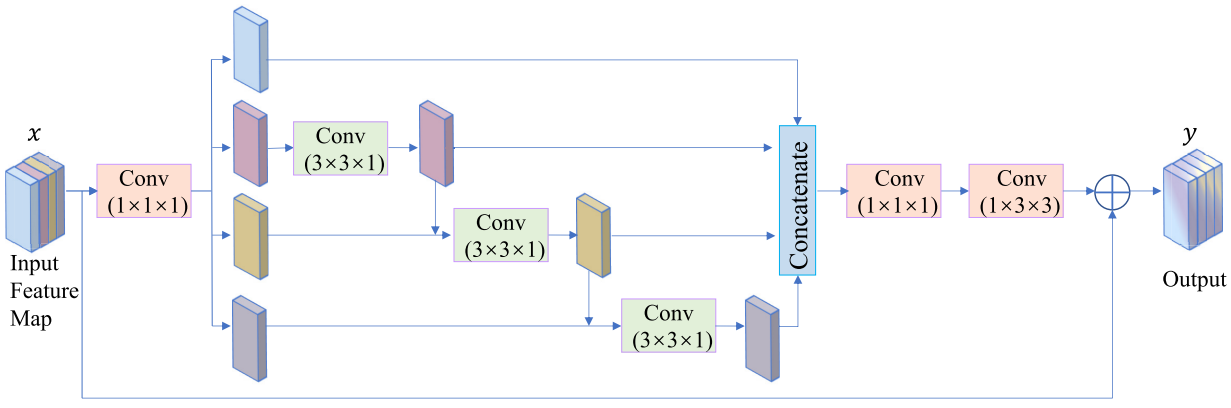


(a) Channel Attention Dynamic Mamba (CADM)

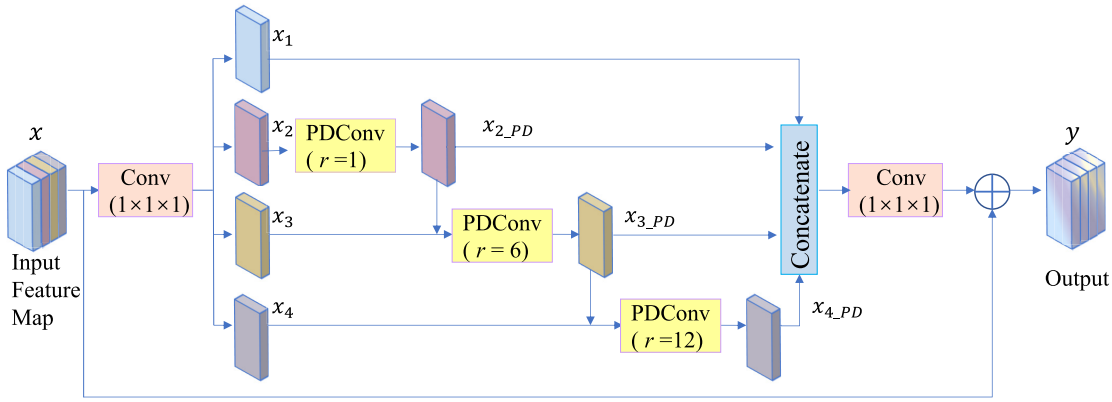


(b) Channel Attention Block for 3D (CAB)

FIGURE 7. The structures of CADM and CAB.



(a) Structure of the HDC module



(b) Structure of the PDHD module

FIGURE 8. Two basic network module structures.

$$
\begin{aligned}
x_{2_PD} &= PDConv^{1}_{3\times3\times3}(x_2),\\
x_{3_PD} &= PDConv^{6}_{3\times3\times3}(x_{2_PD} + x_3),\\
x_{4_PD} &= PDConv^{12}_{3\times3\times3}(x_{3_PD} + x_4)
\end{aligned} \tag{9}
$$

where y denotes the output image and x represents the input image. According to the channel, x is partitioned into four components: x_1, x_2, x_3 and x_4. x_{Cat} represents the

concatenation of the four parts following the processing of the *PDConv*. $PDConv_a^b$ denotes the partial dilation convolution, wherein a indicates the size of the convolution kernels, which is set to $3 \times 3 \times 3$, and b signifies the dilation rate, selected as 1, 6, and 12, respectively. The first branch still employs shortcut connections to preserve the image features through residual learning. PDConv operates across three parallel branches with varying dilation rates, facilitating the interaction of brain tumor feature information between channels through two cascade structures. Then, the outputs of the four branches are concatenated along the channel dimension. Ultimately, it undergoes processing through a $1 \times 1 \times 1$ convolution and is subsequently added to the original input image. The proposed PDConv architecture is illustrated in Figure 9, which is expressed as the following formula:

$$\begin{aligned} y_{PD} &= x_C + x_{PD}, \\ x_{PD} &= Conv_{1\times3\times3}\left(Relu\left(BN\left(Conv_{3\times3\times1}(x_{Cat})\right)\right)\right), \\ x_{Cat} &= \omega_1\left(DConv_{3\times3\times3}(x_p)\right) + \omega_2\left(x_{c-p}\right) \end{aligned} \tag{10}$$

where y_{PD} represents the output image, whereas x_C and x_{PD} signify the input image and the partial dilated convolution of the input image, respectively. The input image with c channels is first divided into two parts: one with p channels and the other with $c - p$ channels. Notably, multi-scale dilated convolution is employed for p channels to capture tumor details. Subsequently, adaptive weight coefficients, ω_1 and ω_2, are allocated to the two components correspondingly, in order to capture the characteristics of various tumors comprehensively. $DConv_{3\times3\times3}$ represents a 3D dilated convolution operation, and BN signifies batch normalization.

This module consists of two core components. First, adopt the scale factor ($p = 4$) to divide the input image into channels, and assign adaptive weights to the two channel groups. On the other hand, multi-scale dilated convolution is introduced to obtain a larger receptive field. Specifically, the $c - p$ channels retain the original input information, while the p channels employ a dilated convolution with a $3 \times 3 \times 3$ kernel size and a dilation rate of γ to emphasize tumor details across various scales with differing receptive fields. This channel fusion strategy mitigates overfitting and enhances feature representation, while the channel-splitting strategy introduces minimal parameter overhead [14].

E. LOSS FUNCTION

We utilize a hybrid loss function combining Dice loss [40] and Cross-Entropy (CE) loss [41], [42]. The Dice loss is formulated as:

$$L_{\text{DICE}} = 1 - \frac{1}{C}\sum_{i=1}^{C}\frac{2\sum_{j=1}^{N}(p_{ij}g_{ij} + \varepsilon)}{\sum_{j=1}^{N}p_{ij} + \sum_{j=1}^{N}(g_{ij} + \varepsilon)} \tag{11}$$

where N is the number of pixels, C represents the number of target classes, p_{ij} denotes the prediction of the i-th class and the j-th voxel, g_{ij} is the binary label, ε is a minimum factor that prevents a zero in the denominator. The CE loss is defined as:

$$L_{CE} = -\sum_{j=1}^{N}\sum_{i=1}^{C} g(j)\cdot\log\big(p(j)\big) \tag{12}$$

where $g(j)$ is a one-hot encoding vector representing the target class of the j-th voxel, $p(j)$ predicts class probabilities for the j-th voxel, and log denotes the natural logarithm. The final loss function is given as:

$$L = 0.5L_{DICE} + 0.5L_{CE} \tag{13}$$

We adopt equal weighting to balance their complementary strengths. Dice loss, being region-based, is particularly effective for handling class imbalance, while CE loss promotes stable voxel-level classification. This strategy does not overemphasize either objective, maintains numerical stability, and eliminates the need for additional hyperparameter tuning.

IV. EXPERIMENTS

This section is divided into three subsections. Subsection IV-A introduces the dataset and evaluation criteria employed for quantitative assessment. Then, Subsection IV-B details the implementation settings. Finally, Subsection IV-C reports the results and analysis, comparing the proposed DM-UNet with existing segmentation models and discussing its effectiveness in brain tumor segmentation.

A. DATASET AND EVALUATION CRITERIA

BraTS2023 [43], [44], [45] dataset comprises a substantial collection of annotated 3D brain MRI scans. Since the validation set lacks ground truth and the test is private, our experiments employ 5-fold cross-validation exclusively on the officially released training set. The dataset comprises 1,251 cases of 3D brain tumor volumes. All MRI images are $240 \times 240 \times 155$ mm^3, and the spacing among voxels is $1 \times 1 \times 1$ mm^3. Each case provides four complementary MRI modalities: T1-weighted (T1), post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 Fluid Attenuated Inversion Recovery (T2-FLAIR). In addition, the corresponding segmentation labels encompass three tumor subregions: WT, TC, and ET.

We adopt the Dice Similarity Coefficient (DSC) and Hausdorff Distance (HD) to evaluate the segmentation performance, which are the most widely adopted evaluation criteria in medical image analysis. DSC represents the overlap rate between the predicted segmentation and the ground truth. It ranges from 0 to 1, and the higher DSC value indicates greater consistency, reflecting more accurate segmentation. In contrast, the HD measures the distance between the boundaries of the predicted and ground truth regions, indicating the spatial accuracy of the segmentation. A lower HD value suggests that the model is more precise in capturing the tumor boundaries. The mathematical

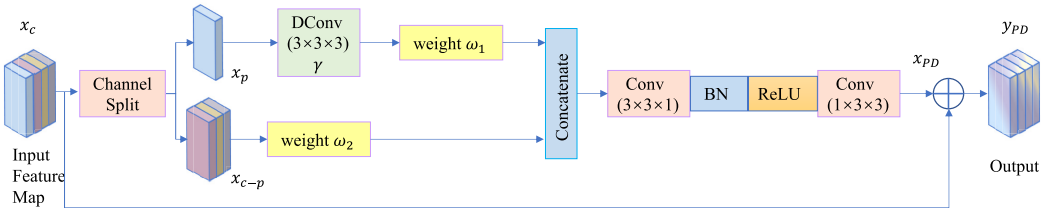


FIGURE 9. The structures of partial dilation convolution (PDConv).

formulations of DSC and HD are presented as follows:

$$\begin{aligned} \text{Dice} &= \frac{2TP}{2TP + FP + FN} \\ \text{Hausdorff} &= \max\{d(A, B),\ d(B, A)\} \end{aligned} \tag{14}$$

where TP denotes the number of true positive voxels, FN corresponds to the number of false negative voxels, and FP indicates the number of false positive voxels. A, B denote the pixel sets of the prediction result and ground truth, respectively. $A = \{a_1, a_2, \ldots, a_n\}$, $B = \{b_1, b_2, \ldots, b_n\}$ are in Euclidean space, and $[d(A, B) = \max_{a\in A} \min_{b\in B} \|a - b\|]$, $[d(B, A) = \max_{b\in B} \min_{a\in A} \|b - a\|]$ are different distances between the surfaces of two objects.

B. IMPLEMENTATION DETAILS

All experiments are implemented using PyTorch 2.5.1 on a workstation equipped with an NVIDIA RTX 3090 GPU. The models are trained for 900 epochs using the AdamW optimizer. The initial learning rate is set 1×10^{-3} and progressively adjusted during training via the cosine annealing scheduling strategy. The training utilizes a random crop size of $128 \times 128 \times 128$ voxels with a batch size of 2. To enhance model generalization and mitigate overfitting, we apply comprehensive data augmentation, including spatial transformations such as random rotations, isotropic scaling, and random mirroring along specified axes. These augmentations are applied uniformly across all four MRI modalities for each case. During inference, a sliding window approach with an overlap rate of 0.5 is adopted. To ensure a fair comparison, all experiments are trained and evaluated under the same settings. Specifically, all methods employed the same preprocessing operations, data augmentation strategies, dataset splits, optimizer settings, learning rate schedules, batch sizes, and training epochs. During the inference process, the same input resolution and evaluation metrics are always used. The results of the competing methods are reproduced using the officially released codes under the same settings.

C. RESULTS AND DISCUSSIONS

This subsection presents a comprehensive evaluation of the proposed DM-UNet model from multiple perspectives. Subsubsection IV-C1 describes the 5-fold cross-validation results, demonstrating the model's stability and generalization performance across different data partitions. Subsubsection IV-C2 provides comparative analyses with state-of-the-art segmentation methods. Finally, Subsubsection IV-C3 reports the ablation studies.

1) 5-FOLD CROSS-VALIDATION

To evaluate the generalization and robustness of our model, we conduct 5-fold cross-validation on the BraTS2023 dataset. The average training time of our model is approximately 4 minutes per epoch, resulting in a total training time of around 60 hours for each fold. In addition, the proposed method achieved an inference latency of 186.52 ms per volume. As shown in Table 2, the results are reported in terms of Dice score and HD for the three tumor subregions. The corresponding mean and standard deviation are compared to assess consistency and stability.

The proposed model achieves an average Dice score of 92.86% for WT, 90.41% for TC, and 85.53% for ET, with an overall average Dice of 89.60% across all folds. In terms of boundary accuracy, the mean HD values are 4.13 mm (WT), 3.91 mm (TC), and 4.31 mm (ET), with an overall average of 4.11 mm. These results demonstrate the model's reliable performance across different tumor subregions.

Among the three subregions, segmentation of the WT yields the highest Dice score and the lowest variability ($\pm$9.48), as WT typically occupies a larger spatial region and presents more continuous anatomical structures. The region of ET exhibits the lowest Dice score (85.53%) and the highest variance ($\pm$20.51), indicating that ET remains the most challenging region due to its irregular shape, small volume, and diverse appearance across patients.

Notably, the performance variation across the five folds remains relatively small, particularly in the WT and TC regions, suggesting that the proposed model maintains stable segmentation ability across different data partitions. The results also confirm that the model achieves a favorable balance between segmentation accuracy and boundary precision, which is essential for clinical applicability in glioma delineation.

2) COMPARISON WITH STATE-OF-THE-ART (SOTA) METHODS

To evaluate the performance of our proposed model, we compare it with several SOTA methods on Fold 0 of the BraTS2023 dataset. These methods include three U-Net-based architectures (HDC-Net [37], ADHDC-Net [13], TDPC-Net [14]), and two Mamba-integrated

TABLE 2. Five-fold cross-validation performance for the BraTS 2023 dataset.

Fold	Dice (%)				HD (mm)			
	WT	TC	ET	Mean	WT	TC	ET	Mean
0	93.11 ± 8.13	91.64 ± 13.02	87.03 ± 18.01	90.59 ± 13.05	3.97 ± 7.49	3.47 ± 7.35	3.78 ± 9.27	3.74 ± 8.04
1	92.79 ± 9.06	90.29 ± 16.59	83.36 ± 24.26	88.81 ± 16.64	4.12 ± 6.74	3.69 ± 7.30	5.13 ± 11.71	4.31 ± 8.58
2	93.19 ± 9.92	91.04 ± 15.73	87.83 ± 16.72	90.69 ± 14.12	3.91 ± 7.97	3.83 ± 7.60	3.88 ± 8.76	3.87 ± 8.11
3	92.15 ± 11.31	89.56 ± 17.46	84.94 ± 21.85	88.88 ± 16.87	4.32 ± 9.03	4.55 ± 12.41	4.52 ± 12.30	4.46 ± 11.25
4	93.04 ± 8.99	89.53 ± 17.15	84.48 ± 21.90	89.02 ± 16.01	4.33 ± 10.43	3.99 ± 8.49	4.24 ± 9.49	4.19 ± 9.47
Mean	92.86 ± 9.48	90.41 ± 15.99	85.53 ± 20.51	89.60 ± 15.33	4.13 ± 8.33	3.91 ± 8.63	4.31 ± 10.31	4.11 ± 9.09

TABLE 3. Comparison of segmentation performance with SOTA methods on BraTS 2023 (Fold=0).

Model	Dice (%)				HD (mm)			
	WT	TC	ET	Mean	WT	TC	ET	Mean
HDC-Net [37]	92.11 ± 11.41	89.01 ± 19.62	84.74 ± 21.42	88.62 ± 17.48	4.84 ± 8.80	4.37 ± 8.44	4.65 ± 10.48	4.62 ± 9.24
ADHDC-Net [13]	92.49 ± 10.44	89.17 ± 19.46	84.71 ± 20.93	88.79 ± 16.94	4.51 ± 8.96	4.81 ± 10.90	4.67 ± 11.11	4.66 ± 10.32
TDPC-Net [14]	92.66 ± 10.15	88.81 ± 20.34	84.48 ± 21.53	88.65 ± 17.34	4.25 ± 7.63	4.61 ± 10.02	4.63 ± 10.46	4.50 ± 9.37
LightM-UNet [34]	92.73 ± 7.95	90.94 ± 15.21	86.78 ± 18.46	90.15 ± 13.87	4.46 ± 8.44	3.13 ± 5.85	3.55 ± 8.54	3.71 ± 7.61
SegMamba [22]	93.59 ± 6.47	91.53 ± 13.78	86.62 ± 18.74	90.58 ± 13.00	3.72 ± 7.04	3.34 ± 6.72	3.83 ± 9.14	3.63 ± 7.63
Ours	93.11 ± 8.13	91.64 ± 13.02	87.03 ± 18.01	90.59 ± 13.05	3.97 ± 7.49	3.47 ± 7.35	3.78 ± 9.27	3.74 ± 8.04

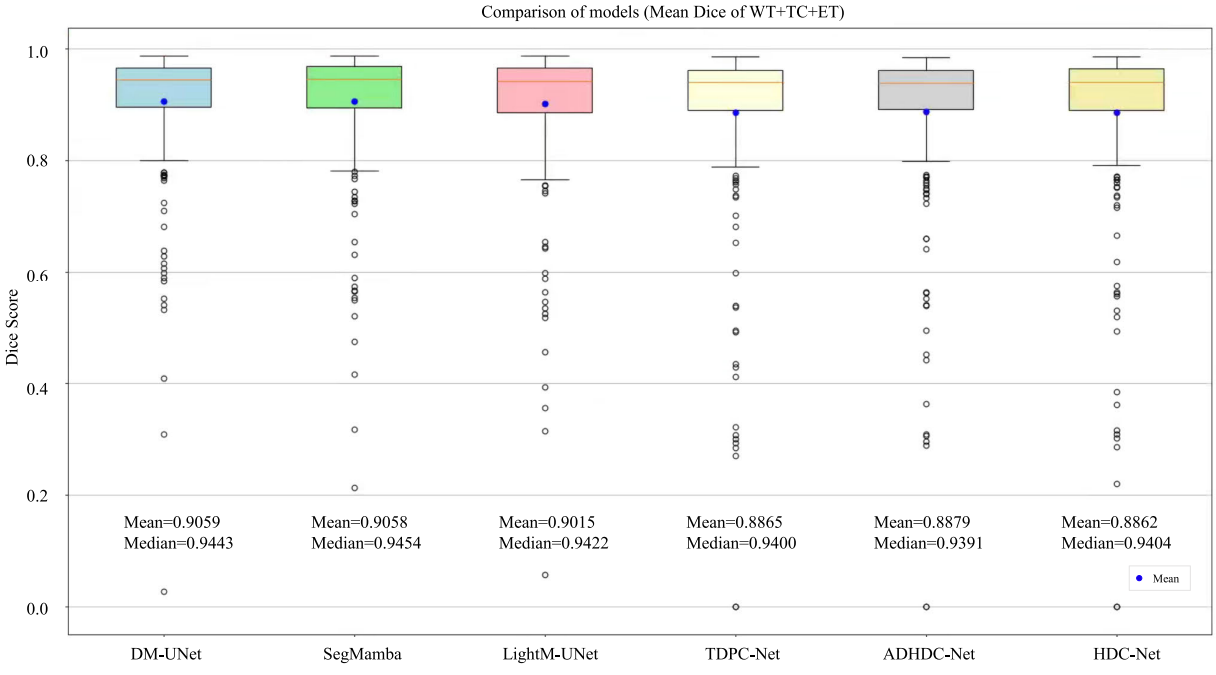


FIGURE 10. The results of the Dice box diagram.

models (LightM-UNet [34], SegMamba [22]). The comparison results are presented in Table 3. Except for SegMamba, the other methods are lightweight methods, and the basic information of each comparison method is shown in Table 4.

Among the U-Net-based methods, ADHDC-Net achieves the highest mean Dice score (88.79%) but suffers from the highest HD mean (4.66 mm), indicating relatively weaker boundary precision. TDPC-Net shows slightly better boundary localization with a lower HD (4.50 mm), but yields lower Dice in ET subregions. Overall, the performance of U-Net variants remains limited, particularly for the more complex and irregular tumor structures such as ET, where Dice scores remain below 85%.

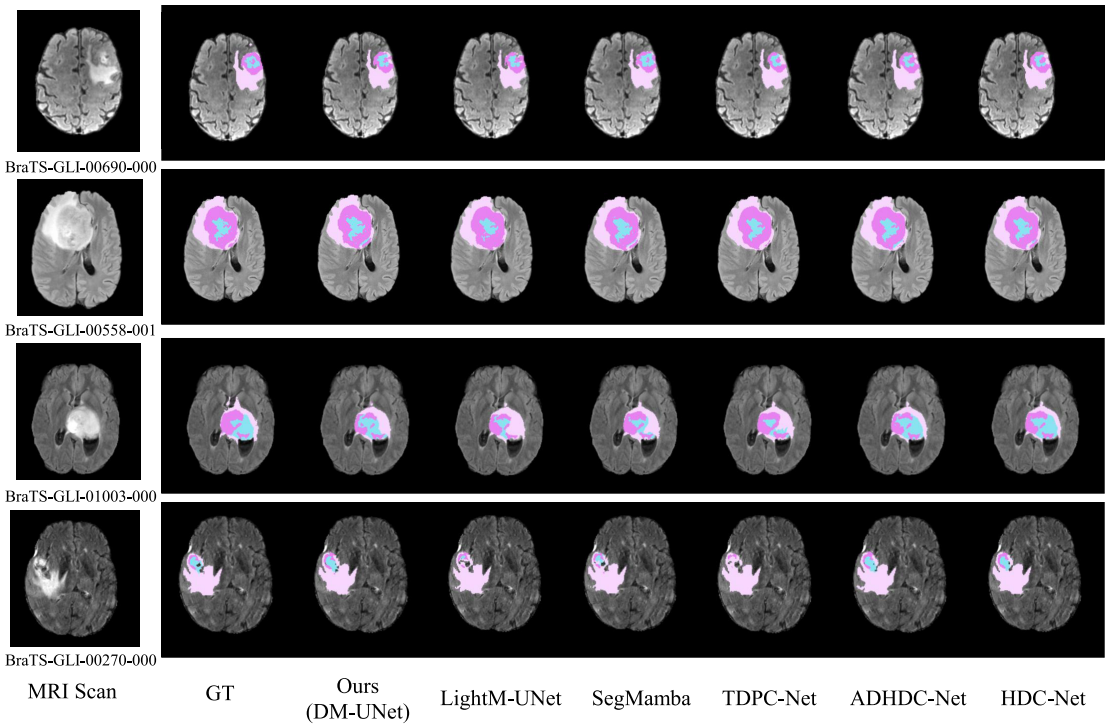


FIGURE 11. 2D visualization comparison of segmentation results from different models.

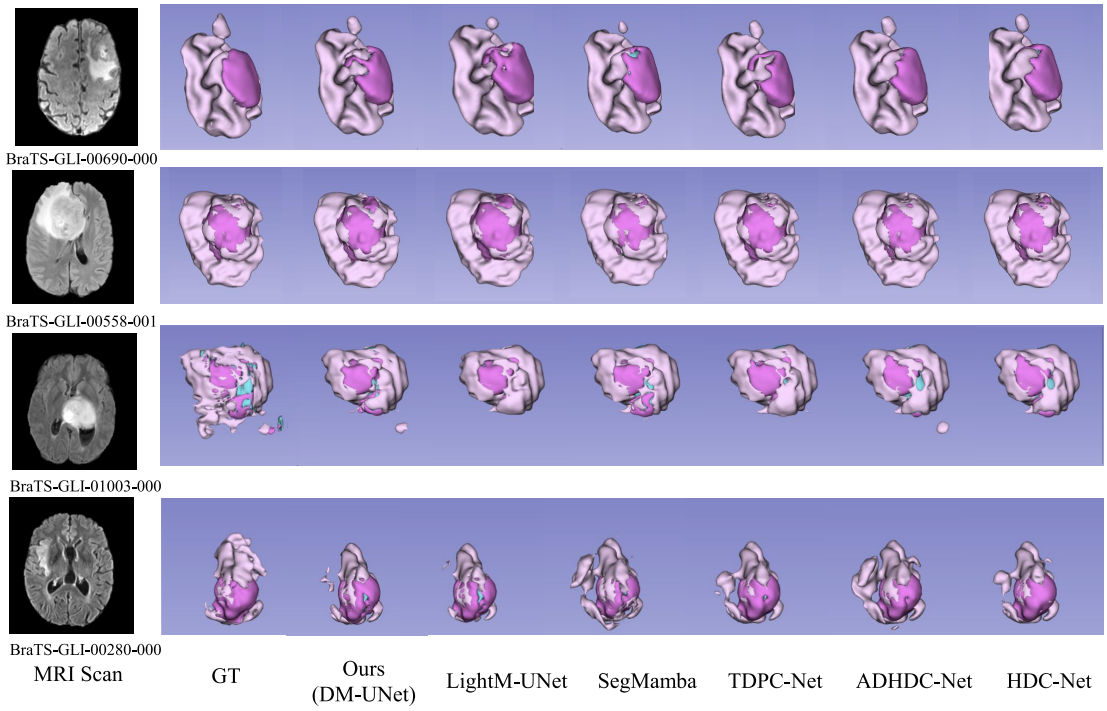


FIGURE 12. 3D visualization comparison of segmentation results from different models.

In contrast, the Mamba-based models demonstrate clear advantages in both accuracy and boundary precision. LightM-UNet improves the mean Dice to 90.15% and reduces HD to 3.71 mm, largely benefiting from its Mamba mechanism. SegMamba performs comparably with a mean Dice of 90.58% and the lowest HD (3.63 mm), showcasing strong boundary alignment capabilities. Our proposed method achieves the highest mean Dice of 90.59%, as well as

TABLE 4. Computational complexity and inference efficiency of the proposed method and SOTA models. All measurements were conducted on NVIDIA RTX 3090 GPU (24GB), batch size=1, input size=(4 × 128 × 128 × 128).

Model	Year	Params (M)	Model Size (MB)	FLOPs (G)	Inference Time (ms)	Memory (MB)
HDC-Net [37]	2021	0.297	1.13	18.78	11.95	413.81
ADHDC-Net [13]	2023	0.297	1.19	19.01	16.65	413.83
TDPC-Net [14]	2025	0.191	0.73	17.37	11.18	837.38
LightM-UNet [34]	2024	0.201	0.92	6.38	160.56	1555.72
SegMamba [22]	2024	67.42	256.97	1460	258.02	3340.51
Ours	2025	**0.093**	**0.36**	14.96	186.52	2066.81

TABLE 5. Paired sample t-test results for segmentation performance comparisons of DM-UNet with other models on the BraTS2023 (Fold=0).

Model 1	Model 2	t Statistic	p Value
DM-UNet	SegMamba	0.0485	0.961394 (>0.05)
DM-UNet	LightM-UNet	2.5320	0.011956 (<0.05)
DM-UNet	TDPC-Net	4.5671	0.000008 (<0.05)
DM-UNet	ADHDC-Net	4.3835	0.000018 (<0.05)
DM-UNet	HDC-Net	4.2934	0.000026 (<0.05)

a consistently low HD (3.74 mm), indicating balanced performance across all tumor subregions. In particular, our model yields the best Dice scores for TC (91.64%) and ET (87.03%), which are the most challenging regions due to their morphological complexity. This suggests that our approach is more effective in capturing fine-grained features and adapting to diverse tumor appearances, likely due to the proposed DTSM.

As shown in Table 4, the proposed method is evaluated in terms of parameter count, model size, FLOPs, inference latency, and memory consumption. Although SegMamba achieves comparable segmentation performance, it contains 67.42M parameters and 1460G FLOPs. In contrast, our model requires fewer than 0.1M parameters, with FLOPs accounting for only approximately 1% of SegMamba. Regarding inference efficiency, Mamba-based methods exhibit higher latency due to SSM's sequential scanning mechanism. Notably, despite this inherent constraint, the proposed model achieves inference latency and memory consumption of 186.52 ms and 2066.81 MB, respectively, outperforming SegMamba by 27.7% and 38.1%. Overall, the proposed method demonstrates a reasonable trade-off between segmentation accuracy and computational overhead.

The segmentation outcomes of DM-UNet and five competing methods are summarized by the box plots in Figure 10. The box represents the interquartile range, while the median, mean, and outliers are indicated by the horizontal line, blue dot, and white circles, respectively. DM-UNet and SegMamba exhibit more compact distributions and shorter whiskers, suggesting better stability and generalization across patients. LightM-UNet remains competitive but achieves slightly lower performance, whereas TDPC-Net, ADHDC-Net, and HDC-Net show wider distributions and more outliers, indicating less consistent segmentation results. The Mamba-based methods demonstrate stronger robustness than the CNN-based counterparts. To evaluate statistical significance, paired sample t-tests were conducted between DM-UNet and the other methods, with the results reported in Table 5. No significant difference is observed between DM-UNet and SegMamba ($p = 0.96$), while DM-UNet significantly outperforms LightM-UNet ($p < 0.05$), TDPC-Net ($p < 0.001$), ADHDC-Net ($p < 0.001$), and HDC-Net ($p < 0.001$), which confirms the competitive and reliable performance of DM-UNet.

To further demonstrate the model's performance, several representative cases are visualized in Figure 11 and Figure 12, presenting 2D and 3D segmentation results. Wherein the whole tumor, tumor core, and enhancing tumor are denoted by pink, purple, and blue regions, respectively. Compared to the other methods, our model produces results that are more consistent with the ground truth, particularly in complex tumor regions. Notably, in the most challenging cases, our method outperforms compared approaches, showing superior accuracy and robustness in brain tumor segmentation.

3) ABLATION STUDIES

a: EFFECTIVENESS OF EACH COMPONENT

To evaluate the effectiveness of each component, we adopt HDC-Net as the baseline and successively integrate two key modules, CADM and PDHD, to assess their contribution to model performance. The results are as shown in Table 6. We find that introducing PDHD reduces the number of parameters by 74.41%, while the Dice score increases by only 1.16%. Integrating CADM leads to a 53.53% reduction in parameters, along with a 1.94% increase in Dice and a 20.78% decrease in HD. When both CADM and PDHD are integrated, the parameters are reduced by 68.69%, the Dice score improves the most by 2.22%, and HD decreases by 19.05%, which is slightly higher than with CADM alone. Overall, the integration of both modules achieves the most significant improvement in segmentation performance, balancing parameter reduction with enhanced accuracy.

b: EFFECTIVENESS OF MAMBA

To evaluate the performance of various Mamba mechanisms, we replaced the DTSM in the CADM block with the bidirectional-scanning from Vision Mamba and the TSMamba from SegMamba. As presented in Table 7, the proposed DTSM achieves the highest Dice score of 90.59%, outperforming TSMamba (90.36%) and bidirectional scanning (89.73%). Correspondingly, DTSM also results in the lowest HD of 3.74 mm, compared to TSMamba (3.82 mm) and bidirectional scanning 4.14 mm. These results demonstrate that DTSM effectively captures discriminative features of tumors with complex morphological variations, thereby improving both segmentation accuracy and boundary delineation.

TABLE 6. Ablation study for modules on the BraTS2023 dataset (Fold=0).

Model	Component		Para (M)	Dice (%)				HD (mm)			
	CADM	PDHD		WT	TC	ET	Mean	WT	TC	ET	Mean
M0			0.297	92.11±11.41	89.01±19.62	84.74±21.42	88.62±17.48	4.84±8.80	4.37±8.44	4.65±10.48	4.62±9.24
M1		✓	0.076	92.01±10.84	90.65±15.44	86.29±18.93	89.65±15.07	4.50±8.07	3.77±7.55	4.14±9.66	4.14±8.43
M2	✓		0.138	93.12±8.88	91.75±12.91	86.17±18.92	90.34±13.57	3.84±6.55	3.29±7.33	3.85±9.69	3.66±7.86
Ours	✓	✓	0.093	93.11±8.13	91.64±13.02	87.03±18.01	90.59±13.05	3.97±7.49	3.47±7.35	3.78±9.27	3.74±8.04

TABLE 7. Ablation study on Mamba scanning strategies (Fold=0).

Scan Orientation	Dice (%)				HD (mm)			
	WT	TC	ET	Mean	WT	TC	ET	Mean
Bidirectional [21]	92.32±10.58	90.58±15.42	86.30±19.04	89.73±15.01	4.30±7.65	3.99±8.50	4.12±9.83	4.14±8.66
TSMamba [22]	92.82±9.21	91.43±14.11	86.83±18.70	90.36±14.01	4.14±7.46	3.52±7.12	3.79±9.06	3.82±7.88
DTSM (Ours)	93.11±8.13	91.64±13.02	87.03±18.01	90.59±13.05	3.97±7.49	3.47±7.35	3.78±9.27	3.74±8.04

TABLE 8. Ablation study on different dilation rates (Fold=0).

Dilation rate	Dice (%)				HD (mm)			
	WT	TC	ET	Mean	WT	TC	ET	Mean
1, 2, 3	92.82±9.29	91.48±14.15	86.91±18.41	90.40±13.95	4.12±7.06	3.29±6.93	3.75±9.22	3.72±7.74
1, 6, 12	93.11±8.13	91.64±13.02	87.03±18.01	90.59±13.05	3.97±7.49	3.47±7.35	3.78±9.27	3.74±8.04
2, 6, 10	93.03±8.76	91.76±13.59	86.81±18.28	90.53±13.54	4.11±8.63	3.13±6.66	3.66±9.21	3.63±8.17

c: EFFECTIVENESS OF DILATED CONVOLUTION

To evaluate the impact of dilation rates on feature extraction, we compare three groups of dilated convolution rates, as shown in Table 8. The second group (rates of 1, 6, and 12) achieved the highest average Dice score (90.59%), indicating better region-wise overlap due to the broader receptive fields. However, it slightly compromised on boundary accuracy, as reflected by a higher HD (3.74 mm). In contrast, the third group (rates of 2, 6, and 10) attained the best HD (3.63 mm), suggesting improved boundary precision. Notably, it also delivered a competitive Dice score of 90.53%. The first group (rates of 1, 2, and 3) had the smallest receptive field and thus showed the weakest performance. This ablation suggests that a moderate dilation spread offers a better trade-off between global context aggregation and local detail preservation.

4) THE LIMITATION AND FUTURE IMPROVEMENT

Figure 13 illustrates four representative failure cases of the model. For each case, the Ground Truth (GT) is shown on the left, while the corresponding predicted segmentation results of the proposed model are presented on the right. These four cases are classified into two types of failures: over-segmentation and under-segmentation.

BraTS-GLI-00454-000: The model exhibited over-segmentation, where additional regions not belonging to the actual tumor are incorrectly included. This error is likely caused by the model's limited ability to distinguish normal brain tissue from lesion boundaries in certain modalities, particularly in areas with similar intensity distributions.

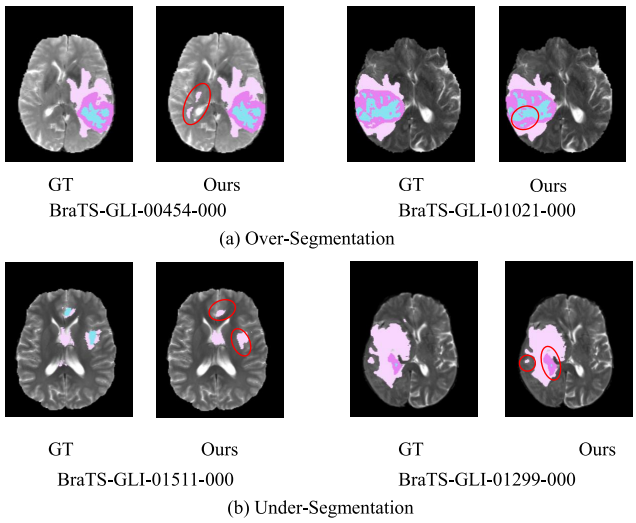


FIGURE 13. Several failure cases.

BraTS-GLI-01021-000: It failed to maintain fine tumor boundaries, with some edge details over-segmented and small tumor edema areas classified as enhancing tumors.

BraTS-GLI-01511-000: The model completely failed to segment the ET region. This failure suggests difficulties in detecting small-volume or low-contrast lesions, potentially due to insufficient representation of similar cases in the training data. BraTS-GLI-01299-000: This case showed a more obvious reduction in the tumor area, with some edge details omitted or blurred. This might arise from the channel

split strategy being overly simplistic, resulting in a failure to achieve a perfect balance between capturing long-range dependencies and fine local structures.

Quantitative analysis indicates that the ET region remains the most challenging sub-region for segmentation. This is reflected by the largest variance in both Dice (±18.01%) and HD (±9.27 mm) among all three subregions. This variability is primarily attributed to the relatively small size of the ET region, where the class imbalance is more severe, making the region more prone to missed or inaccurate segmentation. In terms of computational efficiency, although the proposed framework achieves an extremely compact design with only 0.093M parameters, the sequential scanning mechanism of DTSM introduces a moderately higher inference delay compared to the lightweight alternatives based on CNN, which is a known trade-off inherent to the SSM architecture.

These results highlight the model's limitations in handling small lesions, low-contrast regions, and intricate boundary structures. Future improvement measures include reweighting of small lesion samples during the training process and improving the channel split strategy to enhance the performance of fine segmentation. Additionally, although the proposed framework is extensively evaluated on the BraTS2023 benchmark, its generalization across heterogeneous clinical environments remains to be validated. Subsequent studies will focus on multi-center dataset evaluation to further assess the model's robustness and clinical applicability.

V. CONCLUSION

We propose DM-UNet, a novel lightweight 3D network for brain tumor segmentation, demonstrating that high segmentation performance can be achieved without sacrificing computational efficiency. By integrating DTSM for adaptive directional feature modeling, CADM for global context representation, and PDHD for multi-scale local detail extraction, DM-UNet achieves an effective balance between accuracy and efficiency. Experiments on the BraTS2023 dataset show that DM-UNet has the smallest parameter size among all compared methods while achieving segmentation performance superior to four competing approaches and comparable to SegMamba. These findings suggest that carefully designed lightweight architectures, rather than simply increasing model scale, provide a practical and effective direction for medical image segmentation. Future work will focus on deploying DM-UNet on edge devices and further exploring its feasibility for clinical brain tumor segmentation in resource-constrained environments.

DECLARATIONS

The authors declare no competing interests.

DATA AVAILABILITY

The BraTS 2023 dataset adopted in this study is available from https://www.synapse.org/Synapse:syn51514105.

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CHENGCHENG JIN received the B.Eng. degree in electronic information engineering. She is currently pursuing the Ph.D. degree in computer science with Universiti Sains Malaysia. Her research interests include medical image analysis and deep learning.



NOR SAFIRA ELAINA MOHD NOOR received the B.Eng. degree (Hons.) in biomedical electronic engineering from Universiti Malaysia Perlis (UniMAP), Malaysia, in 2010, the Master of Science degree in electrical and electronic engineering from Universiti Teknologi PETRONAS (UTP), Perak, Malaysia, in 2017, and the Ph.D. degree in computer engineering from Universiti Sains Malaysia (USM), Malaysia, in 2024. She is currently a Lecturer with the Department of Neurosciences, School of Medical Sciences, USM. Her research interests include machine and deep learning applications for brain signals, especially in EEG and MEG and cognitive neuroscience. Recently, she has also been working on analyzing deep brain stimulation (DBS) data for the prognosis of neurological disorders.



HAIDI IBRAHIM (Senior Member, IEEE) received the B.Eng. degree in electrical and electronic engineering from Universiti Sains Malaysia and the Ph.D. degree in image processing from the Centre for Vision, Speech and Signal Processing (CVSSP), University of Surrey, U.K., in 2005. He is currently an Associate Professor at the School of Electrical and Electronic Engineering, Universiti Sains Malaysia. His research interests include digital image and signal processing and analysis.